



NCIE-011

API, Integration & Data Connector Specification

**In development — for human review of full expansion —
Version 1.0**

Specifies detailed API, event, connector and integration-contract architecture beneath the approved NCIE-001–010 foundation: service contracts, API gateway, internal/external/operator/ARGUS APIs, Memory/Agent/Model/Tool/VPF/Evidence service interfaces, connectors, event-bus architecture, schema versioning, correlation, idempotency, async operations, webhooks, error semantics, rate limits, integration security, observability, degraded-integration reconciliation and contract testing.

Integration-only, upstream-consuming: NCIE-011 owns detailed API/event/connector/integration contracts; it does not redefine canonical data, business-domain logic, ARGUS behavior, Memory semantics, Agent/VPF semantics, governance, IAM/security, Evidence meaning, UX, storage or operations.

Expands approved NCIE-011 Skeletal v0.1, itself traced to approved NCIE-001 v1.3, approved NCIE-002 v0.5, approved NCIE-003 v0.4, approved NCIE-004 v1.2, approved NCIE-005 v1.3, approved NCIE-006 v1.2, approved NCIE-007 v1.2, approved NCIE-008 v1.0, approved NCIE-009 v1.0 and approved NCIE-010 v1.0.

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Maintained by: National Communications Intelligence Ecosystem (NCIE)

Document Governance

0.1 Authority Chain

NCIE-001 requirements → NCIE-002 architecture → NCIE-003 canonical data → NCIE-004 engineering baseline → NCIE-005 ARGUS behavioral authority → NCIE-006 Memory & Context architecture → NCIE-007 AI/Agent/Model Orchestration → NCIE-008 Governance, Authority, Accountability & RACI → NCIE-009 Security, IAM & Zero-Trust Architecture → NCIE-010 Evidence, Provenance & Audit Architecture → NCIE-011 API, Integration & Data Connector Specification. NCIE-011 operationalizes the interface obligations every upstream document already establishes; it creates no canonical meaning, institutional authority, security control or Evidence semantics of its own.

0.2 Token-Conservation Convention

Every chapter opens with a short Upstream block citing the specific upstream NCIE-001–010 authority the chapter operationalizes, stating only the interface/contract relationship rather than restating upstream technical, governance, security or Evidence content.

0.3 Governing Integration Invariants Register

These 20 invariants bind every chapter of this document and are stated once here rather than re-derived repeatedly. They are fixed by the approved NCIE-011 Skeletal v0.1 and by the approved NCIE-001–010 authority chain; no institutional decision is required to adopt them. The 12 NCIE-008 governance invariants, the 16 NCIE-009 security invariants, the 19 NCIE-010 Evidence/provenance/audit invariants, and other upstream invariants already binding from NCIE-005/006/007 remain fully in force and are cross-referenced, not restated, in the chapters below.

- API ACCESS ≠ AUTHORIZATION.
- AUTHENTICATION SUCCESS ≠ ACTION AUTHORITY.
- REQUEST ACCEPTED ≠ ACTION COMPLETED.
- HTTP / TRANSPORT SUCCESS ≠ BUSINESS SUCCESS.
- EVENT RECEIVED ≠ AUTHORITATIVE CURRENT STATE.
- EVENT REPLAY ≠ PERMISSION TO REEXECUTE SIDE EFFECTS.
- SCHEMA COMPATIBILITY ≠ SEMANTIC COMPATIBILITY.
- SERVICE CONTRACT ≠ CANONICAL DATA OWNERSHIP.
- INTEGRATION ADAPTER ≠ SYSTEM OF RECORD.
- CONNECTOR CREDENTIAL ≠ UNBOUNDED DATA ACCESS.
- TOOL / CONNECTOR ELIGIBILITY ≠ EXECUTION AUTHORITY.
- CALLER IDENTITY ≠ HUMAN GOVERNANCE ROLE.
- CORRELATION ID ≠ AUTHORIZATION TOKEN.
- RETRY ≠ SAFE REEXECUTION.
- IDEMPOTENCY KEY ≠ BUSINESS UNIQUENESS.
- WEBHOOK DELIVERY ≠ VERIFIED DOWNSTREAM PROCESSING.
- CACHE / REPLICATED RESPONSE ≠ CURRENT AUTHORITATIVE STATE.
- EXTERNAL RESPONSE ≠ INSTITUTIONAL FINDING OR DECISION.
- API VERSIONING ≠ DATA-MODEL VERSIONING.
- INTEGRATION AVAILABILITY ≠ TRUSTWORTHINESS OF CONTENT.

0.4 Integration/Upstream Boundary Note

NCIE-011 owns detailed API, event, connector and integration-contract architecture. It consumes NCIE-003 canonical data meaning, NCIE-005/006/007 ARGUS/Memory/orchestration semantics, NCIE-008 institutional decision rights, NCIE-009 security/IAM controls and NCIE-010 Evidence/provenance/audit semantics, without redefining any of them. Where a matter requires an institutional or engineering choice (a rate limit, a timeout, a partner-onboarding authority) not fixed by an approved source, this document states the contract property and defers the choice to Human Review rather than inventing an answer. Approval of the NCIE-011 Skeletal v0.1 authorized this full expansion only; it did not itself approve any threshold, product/protocol choice or ownership assignment proposed below.

0.5 Human Review Focus Classification

Every chapter's Human Review Focus item carries one of seven labels: RESOLVED BY UPSTREAM BASELINE; PROPOSED DESIGN DEFAULT; INSTITUTIONAL CONFIRMATION REQUIRED; SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED; LEGAL/POLICY CONFIRMATION REQUIRED; ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED; or SOURCE/WORKFLOW DISCOVERY REQUIRED — each additionally marked Blocking or Non-Blocking, and each stating precisely what it blocks. Per directive item 61, a Chapter 31 register item that restates an identical NCIE-008/009 governance or security question cross-references that upstream Human Review ID instead of duplicating it; a new NCIE-011 item is created only for a distinct interface/integration decision. Chapter 31 consolidates every open item, extracted directly from the chapters so the register cannot drift from its sources (mechanically verified per §31.4).

0.6 Status Convention

Skeletal-structure approval authorizes expansion; it does not approve any individual threshold, product/protocol choice or ownership assignment proposed below — those remain IN DEVELOPMENT / FOR HUMAN REVIEW until a separate approval act. Nothing in this document may be cited as an approved institutional decision, a deployed interface, a partner certification, or a completed test.

0.7 Salient Things to Note Discipline

Directive item 75 requires this expansion to programmatically verify an explicit '<N>.0 Salient Things to Note' subsection in all 32 chapters, continuing the discipline established after the NCIE-009 Chapter-27 omission (found by independent human examination and corrected) and reinforced in the NCIE-010 build. Every chapter below renders its Salient Things to Note subsection as an unconditional part of the same build-time helper (NCIE-011's lib.py raises a build error if a chapter's salient content is missing), and Chapter 32's mechanical audit (§32.2) re-verifies this against the final document.

0.8 Document Amendment History

Version: v0.1

Change: Approved skeletal review edition (structural/conceptual review only; full expansion not yet authorized).

Version: v1.0

Change: Full expansion of the approved NCIE-011 Skeletal v0.1 into the complete 32-chapter specification per the NCIE-011 v0.1→v1.0 Claude Full-Expansion Production Directive: API/integration doctrine, seven consolidated matrices, contradiction/outcome-unknown/schema-semantic audits, Human Review Register (Chapter 31) and full traceability/handover (Chapter 32).

Authority: Approved NCIE-001 Master PRD v1.3, approved NCIE-002 System Architecture & Technical Design v0.5, approved NCIE-003 Data Model & Data Dictionary v0.4, approved NCIE-004 Technology Stack & Engineering Blueprint v1.2, approved NCIE-005 ARGUS Intelligence Assistant Specification v1.3, approved NCIE-006 Memory & Context Architecture Specification v1.2, approved NCIE-007 AI, Agent & Model Orchestration Specification v1.2, approved NCIE-008 Governance, Authority, Accountability & RACI Specification v1.0, approved NCIE-009 Security, IAM & Zero-Trust Architecture v1.0, approved NCIE-010 Evidence, Provenance & Audit Architecture v1.0, and the approved NCIE-011 Skeletal v0.1.

Scope: Controlled full expansion only — the approved 32-chapter skeletal structure, chapter purposes and boundaries are preserved exactly; no chapter was added, removed, merged, split or renumbered.

Status: v1.0 IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION. The approved v0.1 skeleton remains historically preserved as the expansion architecture.

Document	NCIE-011 API, Integration & Data Connector Specification
Edition	In development — Version 1.0 (Full Expansion of Approved Skeletal v0.1)
Requirements authority	NCIE-001 Master Product Requirements Document, Version 1.3 — APPROVED

Architecture authority	NCIE-002 System Architecture & Technical Design, Version 0.5 — APPROVED
Canonical data authority	NCIE-003 Data Model & Data Dictionary, Version 0.4 — APPROVED
Engineering baseline	NCIE-004 Technology Stack & Engineering Blueprint, Version 1.2 — APPROVED
ARGUS behavioral authority	NCIE-005 ARGUS Intelligence Assistant Specification, Version 1.3 — APPROVED
Memory & Context authority	NCIE-006 Memory & Context Architecture Specification, Version 1.2 — APPROVED
Orchestration authority	NCIE-007 AI, Agent & Model Orchestration Specification, Version 1.2 — APPROVED
Governance authority	NCIE-008 Governance, Authority, Accountability & RACI Specification, Version 1.0 — APPROVED
Security authority	NCIE-009 Security, IAM & Zero-Trust Architecture, Version 1.0 — APPROVED
Evidence/provenance/audit authority	NCIE-010 Evidence, Provenance & Audit Architecture, Version 1.0 — APPROVED
Prior baseline	NCIE-011 API, Integration & Data Connector Specification, Skeletal v0.1 — APPROVED FOR FULL EXPANSION (historically preserved)
Status	IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION until explicitly approved; the approved v0.1 skeleton remains historically preserved as the expansion architecture

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Mandate, Authority Chain & Integration Boundary

Establish NCIE-011 as the detailed API, event, connector and integration-contract specification within the approved NCIE suite.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: Master Controlled Register v2.0; NCIE-002 Ch.25; NCIE-003/005/009/010 handovers.

1.0 Salient Things to Note

NCIE-011 operationalizes upstream service/interface obligations; it does not redefine product behavior, canonical data semantics, governance authority, security policy, Evidence meaning, UX or storage — Table 1 fixes this boundary so no later chapter can drift into owning what an upstream document already owns.

1.1 Mandate and Scope

NCIE-011 defines the request/response, event, connector and contract-versioning mechanics that operationalize the interface obligations NCIE-001–010 already establish. It does not create product behavior, canonical meaning, institutional authority, security policy, Evidence semantics, UX or storage design.

1.2 Integration Boundary — What NCIE-011 Does Not Own

NCIE-011 does not own canonical data, business-domain logic, ARGUS behavior, Memory semantics, Agent/VPF semantics, institutional governance, security enforcement, Evidence meaning, UX or physical storage. It owns the detailed contract layer connecting all of them.

Table 1. NCIE-011 integration boundary.

Concern	Owned By
Product requirements/scope	NCIE-001
System architecture	NCIE-002
Canonical data model	NCIE-003
Engineering baseline	NCIE-004
ARGUS behavioral authority	NCIE-005
Memory & Context architecture	NCIE-006
AI/Agent/Model orchestration	NCIE-007
Institutional governance, RACI	NCIE-008
Security, IAM, Zero-Trust enforcement	NCIE-009
Evidence lifecycle, provenance, audit	NCIE-010
Detailed API/event/connector/integration contracts	NCIE-011 (this document)
UX, domain business logic, physical storage, operations, detailed testing	NCIE-012/013/014/015/016 (downstream)

1.3 Interfaces to Upstream Authority

Every consequential interface control in this document (authorization enforcement, Evidence mutability, emergency suspension) is expressed as enforcement of an NCIE-008 decision-right, an NCIE-009 security control, or an NCIE-010 Evidence rule, never as new authority invented here.

1.R RACI / Control Ownership

Table 2. Chapter 1 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Adopt/confirm NCIE-011 architecture	Accountable Authority Class (NCIE-008, to be assigned)	Integration Architecture Owner (to be assigned)	NCIE-008/009/010 chapter owners	All institutional role holders

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR11-1-1: Confirm NCIE-011 acceptance/sign-off authority and any unresolved overlap with NCIE-013 domain-module APIs.

Traceability: Master Controlled Register v2.0; NCIE-002 Ch.25; NCIE-003/005/009/010 handovers.

API & Integration Doctrine

Lock the principles governing synchronous, asynchronous, internal and external integration.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 API/event architecture; NCIE-008 governance; NCIE-009 security.

2.0 Salient Things to Note

API transport is not institutional authority; integration success is not business truth; all contracts remain versioned, observable, authorized and semantically bounded — every chapter below is required to restate the relevant piece of this doctrine wherever its subject matter could otherwise be misread as creating authority or truth by default.

2.1 Contract-First Discipline

Every interface is contract-first: explicit ownership, an explicit schema/version reference, and bounded coupling to its consumers. API transport success is not institutional authority (HTTP / TRANSPORT SUCCESS ≠ BUSINESS SUCCESS), and integration availability does not make content trustworthy (INTEGRATION AVAILABILITY ≠ TRUSTWORTHINESS OF CONTENT).

2.2 Fail-Safe Behavior and Semantic Preservation

Interfaces fail safe rather than silently degrading semantics — an unavailable dependency returns an explicit error or pending state, never a fabricated success. Every contract remains versioned, observable, authorized and semantically bounded through to its downstream consumers.

2.R RACI / Control Ownership

Table 3. Chapter 2 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Maintain/interpret the API/integration doctrine	Accountable Authority Class (to be assigned)	Integration Architecture Owner (to be assigned)	NCIE-008/009 owners	All chapter owners

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Non-Blocking) HR11-2-1: Approve any institution-specific integration governance rules not fixed upstream; does not block the doctrine already stated above in the interim.

Traceability: NCIE-002 API/event architecture; NCIE-008 governance; NCIE-009 security.

Service Contract Model & Interface Registry

Define the logical Service Contract and interface-registration model.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.25.5; NCIE-003 canonical data ownership.

3.0 Salient Things to Note

Service Contract records do not create canonical data ownership. Owning service, version and schema references must remain explicit — Tables 4 and 5 fix this explicitly for every interface class defined in this document, so no contract can be read as redefining the data it references.

3.1 Service Contract Identity

A Service Contract record does not create canonical data ownership (SERVICE CONTRACT ≠ CANONICAL DATA OWNERSHIP). Every Service Contract carries an identity, an owning service, a version, request/response schema references, event schema references, a lifecycle/status field, and documentation metadata.

3.2 Interface Catalogue Matrix (directive item 53)

The consolidated catalogue below identifies every interface class defined across this document, its owning service, caller/consumer class, mode, and authorization boundary.

Table 4. Interface Catalogue Matrix.

Interface/Contract	Owning Service	Caller/Consumer Class	Mode	Canonical Schema Reference	Authorization Boundary	Human Review Status
API Gateway entry point (Ch.4)	Gateway	Internal/external callers	Synchronous	N/A (routing layer)	NCIE-009 authn/authz handoff	Holder — Human Review
Internal service APIs (Ch.5)	Owning internal service	Internal services	Synchronous	Service-specific (NCIE-003)	NCIE-009 service-to-service	Holder — Human Review
External partner APIs (Ch.6)	Partner-facing service	External partners	Synchronous	Partner contract version	NCIE-009 external access	Holder — Human Review
Operator/regulatory interfaces (Ch.7)	Operator-facing service	Operators/regulators	Synchronous/async	Governed submission schema	NCIE-009 IAM + Master Register scope	Holder — Human Review

ARGUS API boundary (Ch.8)	ARGUS-facing service	ARGUS	Both	NCIE-005/006 references	NCIE-005 Human-Primary controls	Holder — Human Review
Memory/Context service interfaces (Ch.9)	Memory/Context service	ARGUS, Agents	Synchronous	NCIE-006 v1.2	NCIE-009 access controls	Holder — Human Review
Agent Factory/Runtime APIs (Ch.10)	Agent Factory/Runtime	ARGUS, orchestration	Both	NCIE-003 Agent objects	NCIE-009 Agent security	Holder — Human Review
Model Gateway APIs (Ch.11)	Model Gateway	ARGUS, Agents	Synchronous	NCIE-007 Model Gateway	NCIE-009 model/provider security	Holder — Human Review
Tool Gateway/connector execution APIs (Ch.12)	Tool Gateway	ARGUS, Agents	Synchronous	NCIE-007 Tool Gateway	NCIE-009 Tool/connector security	Holder — Human Review
VPF service interfaces (Ch.13)	VPF service	ARGUS, Agents, services	Synchronous	NCIE-003 VPF canonical data	NCIE-008/009 governance/security	Holder — Human Review
Evidence/custody/audit APIs (Ch.14)	NCIE-010 handover service	Services, ARGUS,	Synchronous	NCIE-010 Ch.3/7/19	NCIE-009 + NCIE-010 rules	Holder — Human Review
Source acquisition/ingestion connectors (Ch.15)	Connector service	External sources	Both	NCIE-010 acquisition provenance	NCIE-009 connector security	Holder — Human Review
Domain data connectors (Ch.16)	Domain connector service	Domain feeds	Both	NCIE-003 domain models	NCIE-009 connector security	Holder — Human Review

Bulk file/batch integration (Ch.17)	Batch service	Internal/external batch sources	Asynchronous	NCIE-003 data lineage	NCIE-009 security	Holder — Human Review
Event bus (Ch.18)	Event bus	All event producers/consumers	Asynchronous	NCIE-002 Ch.25.2-25.5	NCIE-009 event security	Holder — Human Review
Webhook/callback interfaces (Ch.24)	Webhook dispatch service	External subscribers	Asynchronous	Event schema references	NCIE-009 external security	Holder — Human Review

3.3 Service Contract Matrix (directive item 54)

Every Service Contract maps to exactly one owning service and its schema references, without NCIE-011 becoming the canonical owner of the data those schemas describe.

Table 5. Service Contract Matrix.

Service Contract Class	Owning Service	Request/Response Schema Reference	Event Schema Reference	Human Review Status
Internal service contract	Owning internal service (Ch.5)	NCIE-003-derived, service-specific	Domain event taxonomy (Ch.19)	Holder — Human Review
External partner contract	Partner-facing service (Ch.6)	Partner contract version	N/A unless partner subscribes to events (Ch.24)	Holder — Human Review
Operator/regulatory contract	Operator-facing service (Ch.7)	Governed submission schema	Acknowledgement/reconciliation events	Holder — Human Review
Agent/Model/Tool/VPF service contract	NCIE-007 orchestration services (Ch.10-13)	NCIE-003 Agent/VPF canonical objects	Agent/VPF event category (Ch.19)	Holder — Human Review
Evidence/custody/audit contract	NCIE-010 handover service (Ch.14)	NCIE-010 Ch.3/7/19	Evidence/audit event category (Ch.19)	Holder — Human Review

3.R RACI / Control Ownership

Table 6. Chapter 3 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve authority for	Accountable	Service Contract	Owning service	Assurance/Audit

publishing/deprecating service contracts	Authority Class (to be assigned)	Registry Owner (to be assigned)	leads	(NCIE-008)
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Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR11-3-1: Confirm the authority for publishing/deprecating service contracts.

Traceability: NCIE-002 Ch.25.5; NCIE-003 canonical data ownership.

API Gateway & Entry-Point Architecture

Define the governed entry-point model for synchronous APIs.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.25; NCIE-009 Ch.5-6.

4.0 Salient Things to Note

The gateway is an enforcement/mediation boundary, not a source of business authority. Internal/external paths cannot bypass required IAM/security controls — this chapter treats a proposed bypass pattern as a security exception to justify, never a default routing convenience.

4.1 Gateway Is Enforcement, Not Authority

The gateway is an enforcement/mediation boundary — routing, authentication handoff, authorization enforcement, rate/abuse controls, correlation and observability — not a source of business authority. Internal and external paths alike cannot bypass required NCIE-009 IAM/security controls.

4.2 Policy Integration

Gateway policy integration applies current NCIE-009 authorization, classification and DLP rules at the entry point; any permitted internal bypass pattern is a Human Review decision, not a default this document grants.

4.R RACI / Control Ownership

Table 7. Chapter 4 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve internal gateway-bypass patterns	Accountable Authority Class (to be assigned)	Gateway Architecture Owner (to be assigned)	NCIE-009 IAM owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR11-4-1: Confirm any permitted internal bypass patterns, if any, and the approval authority for them.

Traceability: NCIE-002 Ch.25; NCIE-009 Ch.5-6.

Internal Service APIs

Define internal synchronous service-to-service API contract principles.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.25; NCIE-009 service-to-service security.

5.0 Salient Things to Note

Internal does not mean trusted. Service identity, authorization and contract/version discipline apply to every call — this chapter never exempts an internal caller from the same authorization check an external caller would face.

5.1 Internal Does Not Mean Trusted

Service identity, authorization and contract/version discipline apply to every internal call exactly as to an external one. Every internal API carries endpoint ownership, defined request/response patterns, error semantics, a timeout concept, idempotency where applicable, correlation and versioning.

5.2 Stricter Assurance Classes

Which internal API classes require stricter assurance or network/runtime isolation beyond the baseline (e.g., APIs touching Evidence mutability or governance decisions) is a Human Review decision.

5.R RACI / Control Ownership

Table 8. Chapter 5 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve stricter-assurance internal API classes	Accountable Authority Class (to be assigned)	Internal API Owner (to be assigned)	NCIE-009 service-to-service owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR11-5-1: Confirm internal API classes requiring stricter assurance or isolation.

Traceability: NCIE-002 Ch.25; NCIE-009 service-to-service security.

External Partner APIs

Define external/partner API contract and onboarding principles.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.25.6; NCIE-009 external access/security.

6.0 Salient Things to Note

External partner access uses narrowly scoped machine identities and must not broaden institutional data exposure or bypass classification/purpose controls — this chapter's onboarding model binds every partner credential to an explicit scope from the moment it is issued.

6.1 Narrowly Scoped Machine Identity

External partner access uses narrowly scoped machine identities bound to a specific contract version and data-minimization rule; it must not broaden institutional data exposure or bypass classification/purpose controls just because a partner integration exists.

6.2 Onboarding, Throttling, Revocation and Audit

Partner onboarding and offboarding are governed events with an accountable authority; throttling protects service reliability without creating business priority; revocation takes effect immediately; every partner call is audited (NCIE-010 Ch.20-21).

6.R RACI / Control Ownership

Table 9. Chapter 6 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve partner categories/onboarding authority/restricted data classes	Accountable Authority Class (to be assigned)	Partner API Owner (to be assigned)	NCIE-009 external-access owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR11-6-1: Confirm partner categories, onboarding authority and restricted data classes.

Traceability: NCIE-002 Ch.25.6; NCIE-009 external access/security.

Operator / Regulatory Interface Contracts

Define operator-facing integration contracts for regulated or supervised entities.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: Master Register scope; NCIE-001 operator workflows; NCIE-009 IAM.

7.0 Salient Things to Note

Operator interfaces expose only governed data/actions; interface availability or operator identity does not create authority to access every domain object — this chapter's contract scope is the enforced boundary, not the operator's institutional standing.

7.1 Governed Data and Actions Only

Operator interfaces expose only governed data/actions defined by their contract scope; interface availability or operator identity does not itself create authority to access every domain object — the same current-authorization discipline applies as to any other caller.

7.2 Submission, Acknowledgement and Reconciliation

Operator submission/retrieval APIs produce acknowledgements and are reconciled against expected outcomes; operator classes and their authority boundaries by domain are a Human Review decision.

7.R RACI / Control Ownership

Table 10. Chapter 7 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve operator classes/authority boundaries by domain	Accountable Authority Class (to be assigned)	Operator Interface Owner (to be assigned)	NCIE-009 IAM owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [LEGAL/POLICY CONFIRMATION REQUIRED] (Blocking) HR11-7-1: Confirm operator classes and authority boundaries by domain.

Traceability: Master Register scope; NCIE-001 operator workflows; NCIE-009 IAM.

ARGUS API Boundary

Define service interfaces exposed to or invoked by ARGUS without redefining ARGUS behavior.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-005 behavior boundary; NCIE-006 Memory/Context; NCIE-007 orchestration.

8.0 Salient Things to Note

ARGUS may request, retrieve and route governed data/actions but API access does not grant new authority or override Human-Primary controls — every ARGUS-facing interface in this chapter is a pass-through to upstream authority, never a new grant.

8.1 API Access Does Not Grant New Authority

ARGUS may request, retrieve and route governed data/actions through the conversation/session API, Context retrieval interface, Tool request interface and status/event subscriptions — none of these interfaces grants ARGUS new authority or overrides Human-Primary controls already fixed in NCIE-005.

8.2 Degraded/Error Responses

Error and degradation responses are explicit and never silently substituted with a fabricated success; which ARGUS-facing operations are synchronous versus asynchronous is a Human Review decision.

8.R RACI / Control Ownership

Table 11. Chapter 8 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve synchronous/asynchronous boundary for ARGUS-facing operations	Accountable Authority Class (to be assigned)	ARGUS API Owner (to be assigned)	NCIE-005 behavioral owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Non-Blocking) HR11-8-1: Confirm which ARGUS-facing operations may be synchronous versus asynchronous; does not block the interface contracts already defined in the interim.

Traceability: NCIE-005 behavior boundary; NCIE-006 Memory/Context; NCIE-007 orchestration.

Memory & Context Service Interfaces

Define detailed APIs for governed Context assembly, retrieval, injection references and Memory operations.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-006 v1.2; NCIE-009 access controls.

9.0 Salient Things to Note

CONTEXT PROPAGATION ≠ AUTHORIZATION PROPAGATION. API shape must preserve User/Room/Investigation purpose, classification, temporal mode and provenance — this chapter's request/response schemas carry these fields as mandatory, not optional, attributes.

9.1 Propagation Is Not Authorization

CONTEXT PROPAGATION ≠ AUTHORIZATION PROPAGATION. The API shape for every Context request must preserve User/Room/Investigation purpose, classification, temporal mode and provenance exactly as NCIE-006 defines them — a request/response pair carrying Context does not carry the authorization to use it beyond current re-check.

9.2 Write and Invalidation Controls

Write/update, invalidation and expiry are distinct governed operations with their own authorization checks; write authorities and cross-context exception handling (e.g., allowing a bounded exception across investigations) are Human Review decisions.

9.R RACI / Control Ownership

Table 12. Chapter 9 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve Memory/Context write authorities/cross- context exceptions	Accountable Authority Class (to be assigned)	Memory/Context API Owner (to be assigned)	NCIE-006 Memory owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR11-9-1: Confirm write authorities and cross-context exception handling.

Traceability: NCIE-006 v1.2; NCIE-009 access controls.

Agent Factory & Agent Runtime APIs

Define service contracts for Agent discovery, synthesis, Gate A, runtime execution, run status and result packages.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-007 Ch.11-16; NCIE-003 Agent canonical objects; NCIE-009 Agent security.

10.0 Salient Things to Note

Agent Factory API calls cannot self-register Agents or expand permissions. Runtime interfaces must preserve task scope, Agent identity, effective principal and Context bounds — this chapter's API contracts carry these as required fields, not conveniences an integrator may omit.

10.1 No Self-Registration or Authority Expansion Through the API

Agent Factory API calls cannot self-register Agents or expand permissions — every synthesis request, Agent Definition reference and Gate A request/result follows the same self-registration prohibition and current-authorization discipline NCIE-007 already fixes; the API layer adds no exception.

10.2 Runtime Interface Bounds

Run create/status/cancel and result-package retrieval preserve task scope, Agent identity, effective principal and Context bounds exactly as defined upstream; which Agent lifecycle operations require explicit Human/Governance approval via the API is a Human Review decision.

10.R RACI / Control Ownership

Table 13. Chapter 10 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve Agent lifecycle operations requiring explicit approval	Accountable Authority Class (to be assigned)	Agent API Owner (to be assigned)	NCIE-007 Agent Factory owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR11-10-1: Confirm which Agent lifecycle operations require explicit Human/Governance approval.

Traceability: NCIE-007 Ch.11-16; NCIE-003 Agent canonical objects; NCIE-009 Agent security.

Model Gateway APIs

Define service contracts for model routing, invocation and provenance.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-007 Model Gateway; NCIE-009 model/provider security.

11.0 Salient Things to Note

MODEL ELIGIBILITY ≠ DATA DISCLOSURE AUTHORITY. Caller cannot force non-eligible providers or bypass classification/residency rules — this chapter's routing-constraints field is advisory only, never binding on the Model Gateway's own eligibility decision.

11.1 Callers Cannot Force Ineligible Routes

MODEL ELIGIBILITY ≠ DATA DISCLOSURE AUTHORITY. A caller's model request states routing constraints as preferences, not commands — the Model Gateway remains the sole eligibility authority, and no caller can force a non-eligible provider or bypass classification/residency rules through the API.

11.2 Provenance and Fallback Visibility

Every response carries actual model/deployment provenance, fallback status, and usage metadata. Which routing metadata is caller-visible versus internal-only is a Human Review decision.

11.R RACI / Control Ownership

Table 14. Chapter 11 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve caller-visible vs internal routing metadata	Accountable Authority Class (to be assigned)	Model Gateway API Owner (to be assigned)	NCIE-007 Model Gateway owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Non-Blocking) HR11-11-1: Confirm caller-visible versus internal routing metadata; does not block the provenance fields already required in the interim.

Traceability: NCIE-007 Model Gateway; NCIE-009 model/provider security.

Tool Gateway & Connector Execution APIs

Define contracts for Tool invocation, approval binding, execution receipts and outcome reconciliation.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-007 Tool Gateway; NCIE-009 Tool/connector security.

12.0 Salient Things to Note

TOOL ELIGIBILITY ≠ EXECUTION AUTHORITY. REQUEST ACCEPTED ≠ ACTION COMPLETED. OUTCOME UNKNOWN must remain explicit until reconciled — this chapter's execution-receipt schema has no implicit 'assume success' state; every outcome field is either a confirmed result or OUTCOME UNKNOWN.

12.1 Eligibility Is Not Authority, Acceptance Is Not Completion

TOOL ELIGIBILITY ≠ EXECUTION AUTHORITY, and REQUEST ACCEPTED ≠ ACTION COMPLETED. A Tool request carries actor/effective principal, target, and an approval reference where required (NCIE-008 Ch.14); acceptance of the request is not itself proof the action executed.

12.2 Outcome Reconciliation

Execution receipts carry outcome and support reconciliation; where the outcome cannot be confirmed, the API surfaces OUTCOME UNKNOWN explicitly until reconciled — never silently resolved to success. High-risk Tool classes and their approval-reference requirements are a Human Review decision.

12.R RACI / Control Ownership

Table 15. Chapter 12 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve high-risk Tool classes/approval-reference requirements	Accountable Authority Class (to be assigned)	Tool Gateway API Owner (to be assigned)	NCIE-009 Tool/connector-security owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR11-12-1: Confirm high-risk Tool classes and approval-reference requirements.

Traceability: NCIE-007 Tool Gateway; NCIE-009 Tool/connector security.

VPF Service Interfaces

Define API contracts for profile lookup, validation request, Validation Artifact retrieval and status/revalidation operations.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003 VPF canonical data; NCIE-007 Ch.26; NCIE-008/009 governance/security.

13.0 Salient Things to Note

VPF PASS ≠ HUMAN APPROVAL and VPF PASS ≠ EVIDENCE. API responses must preserve exact validation disposition and historical artifact identity — this chapter's response schema names each disposition explicitly rather than reducing it to a pass/fail flag.

13.1 Preserve Exact Disposition

VPF PASS ≠ HUMAN APPROVAL and VPF PASS ≠ EVIDENCE. Every API response preserves the exact validation disposition (PASS/QUALIFIED/REJECT/HUMAN_REVIEW_REQUIRED/VALIDATION_UNAVAILABLE) and the historical Validation Artifact identity — the API layer never collapses these into a simplified boolean.

13.2 Callable Principals

Profile lookup, validation request, artifact retrieval and history lookup are each distinct operations; which technical principals may call which VPF operations is a Human Review decision.

13.R RACI / Control Ownership

Table 16. Chapter 13 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve VPF-operation caller eligibility by principal class	Accountable Authority Class (to be assigned)	VPF API Owner (to be assigned)	NCIE-007 VPF architecture owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR11-13-1: Confirm which VPF operations are callable by which technical principals.

Traceability: NCIE-003 VPF canonical data; NCIE-007 Ch.26; NCIE-008/009 governance/security.

Evidence, Custody & Audit APIs

Define detailed interface contracts handed off by NCIE-010.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-010 Ch.3, 7, 19 and downstream handover.

14.0 Salient Things to Note

Evidence Object, custody-event and audit-event interfaces must preserve canonical identity, provenance, authorization, classification and historical integrity; API design cannot redefine Evidence semantics — every operation in this chapter maps one-to-one to an NCIE-010 lifecycle event, never a new one.

14.1 API Design Cannot Redefine Evidence Semantics

Evidence Object, custody-event and audit-event interfaces preserve canonical identity, provenance, authorization, classification and historical integrity exactly as NCIE-010 defines them — the API layer is a transport for those semantics, never a reinterpretation of them.

14.2 Mutability and Privileged Access

Register/retrieve, custody-event append, integrity verification, annotation, version-state and audit-event append/query operations follow NCIE-010's append-only and correction-not-erasure rules. Mutability expectations and privileged audit access are a Human Review decision.

14.R RACI / Control Ownership

Table 17. Chapter 14 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve mutability expectations/privileged audit access	Accountable Authority Class (to be assigned)	Evidence API Owner (to be assigned)	NCIE-010 Evidence architecture owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR11-14-1: Confirm mutability/append-only expectations and privileged audit access.

Traceability: NCIE-010 Ch.3, 7, 19 and downstream handover.

Source Acquisition & Ingestion Connectors

Define connector contracts for governed source acquisition and ingestion.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-001 acquisition; NCIE-002 source/event architecture; NCIE-010 acquisition provenance.

15.0 Salient Things to Note

Connector success does not make source material Evidence or canonical truth. Acquisition provenance and source health must remain explicit — Table 18 fixes a provenance and security obligation for every connector class so none can bypass NCIE-010 qualification simply by succeeding at the transport layer.

15.1 Connector Success Is Not Evidence

A successful pull/push connector call does not make source material Evidence or canonical truth (SOURCE DATA ≠ EVIDENCE, inherited from NCIE-010) — it produces a source reference that still requires NCIE-010 Chapter 4 registration and qualification.

15.2 Health, Provenance and Quarantine

Every connector records source identity, the acquisition request, validation results, connector health, retry/checkpoint state, and acquisition provenance; material that fails validation is quarantined rather than silently discarded or silently promoted.

15.3 Connector Matrix (directive item 57)

The consolidated matrix below maps connector classes to identity, scope, security and reconciliation requirements without inventing institutional actors.

Table 18. Connector Matrix.

Connector Class	Identity	Scope	Schema Mapping	Provenance	Security	Health/Checkpoint	Human Review Status
Source acquisition/ingestion (Ch.15)	Connector service identity	Named source system(s)	Source-specific to NCIE-003 canonical model	NCIE-010 acquisition provenance	NCIE-009 connector credential (scoped)	Retry/checkpoint per source	Holder — Human Review
Domain data connector (Ch.16)	Domain connector service identity	Named domain feed	NCIE-003 domain data model	NCIE-010 acquisition provenance	NCIE-009 connector credential (scoped)	Checkpoint per feed batch	Holder — Human Review
Bulk file/batch (Ch.17)	Batch service identity	Named file/batch source	Manifest-declared schema version	NCIE-010 acquisition provenance	NCIE-009 transfer security	Transfer status per manifest	Holder — Human Review
External partner connector (Ch.6)	Partner machine identity	Partner contract scope	Partner contract schema	NCIE-010 acquisition provenance	NCIE-009 external access (narrowly)	Health per partner endpoint	Holder — Human Review

scoped)

15.R RACI / Control Ownership

Table 19. Chapter 15 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve connector classes requiring manual attestation	Accountable Authority Class (to be assigned)	Connector Architecture Owner (to be assigned)	NCIE-010 acquisition owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR11-15-1: Confirm connector classes requiring manual attestation or source-specific approval.

Traceability: NCIE-001 acquisition; NCIE-002 source/event architecture; NCIE-010 acquisition provenance.

Domain Data Connectors

Define reusable patterns for Network, Traffic, Revenue, Mobile Money, SIM, Anti-Fraud, REWS and related domain feeds without preempting NCIE-013 business logic.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-013 downstream functional modules; NCIE-003 domain data models.

16.0 Salient Things to Note

Connector specification handles transport/contracts, not domain interpretation or institutional Findings — this chapter's schema-mapping and validation fields never encode a business rule that belongs to NCIE-013.

16.1 Transport and Contracts, Not Interpretation

This chapter's connector specification handles transport, schema mapping, validation, checkpointing, error handling and event emission for domain feeds — it does not interpret domain meaning or produce institutional Findings; that remains NCIE-013's business logic.

16.2 Standard vs Bespoke Connectors

Which domain connectors are standard, reusable patterns versus bespoke, feed-specific integrations is a Human Review decision, informed by NCIE-013's actual functional-module requirements.

16.R RACI / Control Ownership

Table 20. Chapter 16 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve standard vs bespoke domain connector classification	Accountable Authority Class (to be assigned)	Domain Connector Owner (to be assigned)	NCIE-013 domain-module owner (downstream)	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Non-Blocking) HR11-16-1: Confirm which domain connectors are standard versus bespoke; does not block the reusable pattern already defined in the interim.

Traceability: NCIE-013 downstream functional modules; NCIE-003 domain data models.

Bulk File & Batch Integration

Define batch/file ingestion and export contracts.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-001 bulk acquisition; NCIE-003 data lineage; NCIE-009 security.

17.0 Salient Things to Note

Bulk transfer requires the same authorization, provenance, classification and integrity rules as APIs; file availability is not successful processing — this chapter's transfer-status field distinguishes 'delivered' from 'validated and processed' as separate states.

17.1 Same Rules as APIs

Bulk transfer requires the same authorization, provenance, classification and integrity rules as synchronous APIs; file availability on a transfer endpoint is not successful processing (REQUEST ACCEPTED ≠ ACTION COMPLETED applies equally to batch).

17.2 Manifest and Partial Failure

Every transfer carries a manifest, file identity, checksum reference, schema version, transfer status, validation result, and reconciliation of any partial failure; accepted transfer classes/formats and size/volume thresholds are Human Review decisions, not invented here.

17.R RACI / Control Ownership

Table 21. Chapter 17 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve accepted transfer classes/formats/thresholds	Accountable Authority Class (to be assigned)	Batch Integration Owner (to be assigned)	NCIE-004 engineering owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR11-17-1: Confirm accepted transfer classes/formats and size/volume thresholds.

Traceability: NCIE-001 bulk acquisition; NCIE-003 data lineage; NCIE-009 security.

Event Bus Architecture & Event Contract Model

Define asynchronous event contracts and event-bus behavior.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.25.2-25.5.

18.0 Salient Things to Note

EVENT RECEIVED ≠ AUTHORITATIVE CURRENT STATE. Consumers must reconcile current state where required; events remain versioned, traceable and idempotent-consumer safe — Table 22's final column fixes what a consumer must reconcile against for every event category, so 'I received the event' is never treated as 'I know the current state.'

18.1 Received Is Not Authoritative

EVENT RECEIVED ≠ AUTHORITATIVE CURRENT STATE. A consumer that receives an event must reconcile against current source state where the downstream decision requires it — the event itself is a notification, not a standing proof of present truth.

18.2 Event Contract Fields

Every event carries identity/type, schema version, correlation ID, publisher, timestamp, a payload or reference, an ordering key where applicable, delivery semantics, and stated consumer responsibilities.

18.3 Event Contract Matrix (directive item 56)

The consolidated matrix below maps event categories to publisher, delivery semantics and reconciliation requirements.

Table 22. Event Contract Matrix.

Event Category	Publisher	Typical Consumers	Delivery Semantics	Ordering/Replay	Authoritative-State Reconciliation
Source events (Ch.19)	Acquisition/ingestion connectors	Domain services, Evidence pipeline	At-least-once (Human Review to confirm exactly-once needs)	Per-source ordering key	Reconcile against NCIE-010 Evidence state
KPI events (Ch.19)	Domain analytics services	Dashboards, ARGUS	At-least-once	Per-metric ordering key	Reconcile against source-of-record metric store
Situation/Alert events (Ch.19)	Detection/analytics services	ARGUS, workflow engine	At-least-once, low-latency preferred	Per-situation ordering key	Reconcile against situation record current status
Workflow events (Ch.19)	Workflow engine	ARGUS, downstream services	At-least-once	Per-workflow-instance ordering key	Reconcile against workflow instance current state
Collaboration events (Ch.19)	Collaboration service	ARGUS, notification service	At-least-once	Per-thread ordering key	Reconcile against collaboration

Agent/VPF events (Ch.19)	NCIE-007 orchestration services	Observability, audit,	At-least-once	Per-Run ordering key	record Reconcile against NCIE-007 Run/VPF state
Evidence/audit events (Ch.19)	NCIE-010 handover service	Audit consumers, compliance	At-least-once, append-only	Per-Evidence-Object ordering key	Reconcile against NCIE-010 Evidence/audit record
Security events (Ch.19)	NCIE-009 security services	SOC/monitoring, incident response	At-least-once, low-latency preferred	Per-incident ordering key	Reconcile against NCIE-009 security state

18.R RACI / Control Ownership

Table 23. Chapter 18 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve event-retention/replay policy by category	Accountable Authority Class (to be assigned)	Event Bus Architecture Owner (to be assigned)	Reviewer	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR11-18-1: Confirm event-retention/replay policy and category-specific delivery expectations.

Traceability: NCIE-002 Ch.25.2-25.5.

Event Categories & Domain Event Taxonomy

Define the event taxonomy across source, PM/KPI, Alert/Situation, workflow, collaboration, Agent/VPF, Evidence and security domains.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 event categories; NCIE-003/005/007/009/010 downstream events.

19.0 Salient Things to Note

Event categories must trace to owning domain semantics; event publication does not transfer ownership of the underlying business state — this chapter's taxonomy is a routing classification, never a claim of ownership over what it classifies.

19.1 Publication Does Not Transfer Ownership

Event categories trace to their owning domain's semantics (Table 22, Chapter 18); publishing an event does not transfer ownership of the underlying business state to the event bus or to any consumer — the owning domain service remains the source of record.

19.2 Additional Categories

Any additional event category beyond the eight defined in Table 22, and its ownership assignment, is a Human Review decision.

19.R RACI / Control Ownership

Table 24. Chapter 19 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve additional event categories and ownership	Accountable Authority Class (to be assigned)	Event Taxonomy Owner (to be assigned)	Domain service owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Non-Blocking) HR11-19-1: Approve any additional event categories and ownership; does not block the eight-category baseline already defined in the interim.

Traceability: NCIE-002 event categories; NCIE-003/005/007/009/010 downstream events.

Schema Registry & Contract Versioning

Define schema/version governance for APIs and events.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.25.4; NCIE-003 versioned canonical data.

20.0 Salient Things to Note

SCHEMA COMPATIBILITY ≠ SEMANTIC COMPATIBILITY. Additive-first evolution is preferred; breaking changes require a new major contract version rather than silent in-place redefinition — this chapter treats a passed compatibility check as necessary, never sufficient, evidence a change is safe.

20.1 Compatibility Is Not Semantic Equivalence

SCHEMA COMPATIBILITY ≠ SEMANTIC COMPATIBILITY. A schema-compatible change (e.g., an additive field) can still change business meaning; passing schema validation never substitutes for reviewing the actual semantic impact of a change.

20.2 Additive-First Evolution

Additive-first evolution is preferred; a breaking change requires a new major contract version rather than silent in-place redefinition (API VERSIONING ≠ DATA-MODEL VERSIONING — a contract version change does not itself imply or require a canonical data-model change). Deprecation windows and compatibility authority are Human Review decisions.

20.R RACI / Control Ownership

Table 25. Chapter 20 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve deprecation windows/compatibility authority	Accountable Authority Class (to be assigned)	Schema Registry Owner (to be assigned)	NCIE-003 canonical data owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR11-20-1: Confirm deprecation windows and compatibility authority.

Traceability: NCIE-002 Ch.25.4; NCIE-003 versioned canonical data.

Correlation, Causation & Trace Propagation

Define correlation and trace identifiers across API, event, Tool, Agent, Evidence and workflow paths.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 correlation requirement; NCIE-003 provenance; NCIE-010 audit/provenance.

21.0 Salient Things to Note

CORRELATION ID ≠ AUTHORIZATION TOKEN and correlation does not establish causality or institutional responsibility by itself — this chapter's correlation ID is never accepted anywhere in this document as a substitute for an authorization check.

21.1 Correlation Is Not Authorization or Causality

CORRELATION ID ≠ AUTHORIZATION TOKEN. A correlation ID lets related requests/events be traced together; it carries no authorization weight and does not itself establish causality or institutional responsibility (LINEAGE ≠ CAUSALITY, inherited from NCIE-010).

21.2 Trace Propagation Limits

Correlation ID, causation reference, trace context, parent/child request linkage, workflow/Run IDs and audit references propagate across every layer this document defines. Cross-boundary trace propagation limits for privacy/security (e.g., not propagating an internal trace ID to an external partner) are a Human Review decision.

21.R RACI / Control Ownership

Table 26. Chapter 21 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve cross-boundary trace-propagation limits	Accountable Authority Class (to be assigned)	Correlation/Trace Owner (to be assigned)	NCIE-009 privacy/security owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR11-21-1: Confirm cross-boundary trace propagation limits for privacy/security.

Traceability: NCIE-002 correlation requirement; NCIE-003 provenance; NCIE-010 audit/provenance.

Idempotency, Duplicate Suppression & Safe Retry

Define safe retry semantics for synchronous and asynchronous integration.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 reliability; NCIE-007 Tool idempotency/outcome semantics.

22.0 Salient Things to Note

RETRY ≠ SAFE REEXECUTION. IDEMPOTENCY KEY ≠ BUSINESS UNIQUENESS. Duplicate suppression must not hide legitimate repeated business actions — this chapter never treats the presence of an idempotency key alone as sufficient proof a retry is safe to reexecute.

22.1 Retry Is Not Automatically Safe

RETRY ≠ SAFE REEXECUTION, and IDEMPOTENCY KEY ≠ BUSINESS UNIQUENESS. A retry is only safe where an idempotency key, replay detection and side-effect-safety design make it so; an idempotency key deduplicates a specific request attempt, it does not itself establish that the underlying business object is unique.

22.2 Duplicate Suppression Does Not Hide Legitimate Repeats

Duplicate suppression targets accidental retransmission, not legitimate repeated business actions (e.g., two genuinely separate submissions that happen to look similar) — which operation classes require mandatory idempotency keys is a Human Review decision.

22.R RACI / Control Ownership

Table 27. Chapter 22 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve mandatory-idempotency-key operation classes	Accountable Authority Class (to be assigned)	Idempotency/Retry Owner (to be assigned)	NCIE-007 Tool Gateway owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR11-22-1: Confirm which operation classes require mandatory idempotency keys.

Traceability: NCIE-002 reliability; NCIE-007 Tool idempotency/outcome semantics.

Asynchronous Operations & Long-Running Workflows

Define contracts for jobs/runs that cannot complete synchronously.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 workflow states; NCIE-004/007 runtime patterns.

23.0 Salient Things to Note

REQUEST ACCEPTED ≠ ACTION COMPLETED. APIs must expose accepted, running, waiting, failed, cancelled, completed and outcome-unknown states where applicable — Table 28 fixes exactly these seven states and their valid transitions so no operation resource can skip a state or silently resolve Outcome Unknown to Completed.

23.1 Accepted Is Not Completed

REQUEST ACCEPTED ≠ ACTION COMPLETED. Every long-running operation is represented as a resource with an explicit status, polling or subscription access, cancellation where supported, progress reporting, a timeout concept, and a completion receipt — never as a bare 202 with no further visibility.

23.2 Asynchronous-State Matrix (directive item 59)

The state model below is consistent across every long-running operation this document defines, without conflicting with upstream workflow/Agent/Tool state models.

Table 28. Asynchronous-State Matrix.

State	Meaning	Valid Transitions	Caller-Visible Receipt
Accepted	Request validated and queued; execution not yet started	→ Running, → Cancelled	Acceptance receipt with operation ID
Running	Execution in progress	→ Waiting, → Completed, → Failed, → Cancelled, → Outcome Unknown	Progress receipt (where available)
Waiting	Execution paused pending an external dependency (e.g., approval, Tool outcome)	→ Running, → Cancelled, → Failed	Waiting-reason receipt
Cancelled	Terminated by caller or governed authority before completion	Terminal	Cancellation receipt with reason
Failed	Terminated due to an identified failure	Terminal	Failure receipt with error taxonomy reference (Ch.25)
Outcome Unknown	Terminal state could not be confirmed (e.g., after a timeout or infrastructure gap)	→ Reconciled (via Ch.29 reconciliation, not a caller-visible transition)	Outcome-unknown receipt, never silently resolved to Completed
Completed	Execution finished with a confirmed result	Terminal	Completion receipt with result reference

23.R RACI / Control Ownership

Table 29. Chapter 23 RACI / control ownership.

Function	Accountable	Responsible	Consulted	Informed
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	(NCIE-008)	(Integration)		
Approve status- vocabulary extensions/cancellation guarantees	Accountable Authority Class (to be assigned)	Async Operations Owner (to be assigned)	Reviewer	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Non-Blocking) HR11-23-1: Confirm status vocabulary extensions and cancellation guarantees; does not block the seven-state baseline already defined in Table 28.
- Traceability: NCIE-002 workflow states; NCIE-004/007 runtime patterns.*

Webhook & Callback Interfaces

Define outbound callback/webhook behavior for external integrations.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 event architecture; NCIE-009 external security.

24.0 Salient Things to Note

WEBHOOK DELIVERY ≠ VERIFIED DOWNSTREAM PROCESSING. Delivery status, signature/authentication, retry and consumer acknowledgement remain distinct — this chapter never treats a 200 response from a subscriber's endpoint as proof the event was correctly processed.

24.1 Delivery Is Not Verified Processing

WEBHOOK DELIVERY ≠ VERIFIED DOWNSTREAM PROCESSING. A successful HTTP delivery to a subscriber's endpoint confirms transport receipt, not that the subscriber correctly processed the payload — delivery status, signature/authentication, retry and consumer acknowledgement remain four distinct, separately tracked concepts.

24.2 Subscription Lifecycle

Subscription, endpoint validation, event filtering, delivery attempts, retries, acknowledgement, disablement and audit are each governed events. Allowed webhook consumers and retry/deactivation policy are Human Review decisions.

24.R RACI / Control Ownership

Table 30. Chapter 24 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve webhook consumer eligibility/retry-deactivation policy	Accountable Authority Class (to be assigned)	Webhook Owner (to be assigned)	NCIE-009 external-security owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR11-24-1: Confirm allowed webhook consumers and retry/deactivation policy.

Traceability: NCIE-002 event architecture; NCIE-009 external security.

Error Semantics & Failure Taxonomy

Define consistent API/event error behavior across NCIE services.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 failure modes; NCIE-005 degraded behavior; NCIE-007 Tool/VPF states.

25.0 Salient Things to Note

Errors must distinguish authentication, authorization, validation, conflict, unavailable dependency, timeout, accepted-but-pending, outcome unknown and internal failure. Failures must not be collapsed into generic success/failure — Table 31 fixes nine distinct classes each with its own retryability and message-exposure rule.

25.1 Failures Are Not Collapsed

External response status is not itself an institutional Finding or Decision (EXTERNAL RESPONSE ≠ INSTITUTIONAL FINDING OR DECISION). Every error distinguishes authentication, authorization, validation, conflict, unavailable-dependency, timeout, accepted-but-pending, outcome-unknown and internal-failure classes — none may be collapsed into a generic success/failure boolean.

25.2 Error-State Matrix (directive item 58)

The matrix below fixes retryability and caller-safe messaging per error class without exposing sensitive internal details by default.

Table 31. Error-State Matrix.

Error Class	Meaning	Retryable?	Caller-Safe Message Discipline
Authentication failure	Caller identity not established	No (caller must re-authenticate)	Generic auth-failure message; no internal detail
Authorization failure	Identity established, action not currently authorized	No	Generic denial message; does not reveal why access was denied by default
Validation failure	Request malformed or fails schema/business validation	No (until corrected)	Machine-readable field-level detail; no internal system detail
Conflict	Request conflicts with current state (e.g., version mismatch)	Conditionally (after refetch)	Current-state reference; no internal detail
Unavailable dependency	A required upstream dependency is down or degraded	Yes (with backoff)	Generic unavailability message; retry-after guidance
Timeout	No response within the operation's bound	Conditionally (idempotency-key-gated)	Generic timeout message; correlation reference for follow-up
Accepted but pending	Request accepted; result not yet available	N/A (poll/subscribe instead)	Operation-status reference, not an error in the ordinary sense
Outcome unknown	Terminal state could not be confirmed	No (requires reconciliation, not blind retry)	Explicit outcome-unknown message; never silently resolved

Internal failure	Unexpected internal error	Conditionally (idempotency-key-gated)	Generic internal-error message; no stack trace or internal system detail by default
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25.R RACI / Control Ownership

Table 32. Chapter 25 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve public vs internal error-detail exposure	Accountable Authority Class (to be assigned)	Error Taxonomy Owner (to be assigned)	NCIE-009 security owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR11-25-1: Confirm public versus internal error-detail exposure.

Traceability: NCIE-002 failure modes; NCIE-005 degraded behavior; NCIE-007 Tool/VPF states.

Rate Limits, Quotas & Backpressure

Define interface-level load protection without inventing unsupported numeric thresholds.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-002 NFRs; NCIE-004 engineering; NCIE-009 abuse controls.

26.0 Salient Things to Note

Rate limiting protects service reliability; it does not create business priority or institutional authority — this chapter never lets a caller's rate-limit tier be cited as evidence of its institutional standing.

26.1 Reliability, Not Business Priority

Rate limiting protects service reliability; it does not create business priority or institutional authority — a caller that has not been rate-limited is not thereby more authorized or more important than one that has.

26.2 Backpressure Concepts

Per-principal limits, burst control, queue/backpressure behavior, retry-after semantics, event-consumer lag and overload behavior are defined conceptually; specific numeric limits, quotas and SLOs are not invented here. Quota ownership and high-priority exception governance are Human Review decisions.

26.R RACI / Control Ownership

Table 33. Chapter 26 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve quota ownership/high-priority exception governance	Accountable Authority Class (to be assigned)	Rate-Limit/Backpressure Owner (to be assigned)	NCIE-009 abuse-control owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR11-26-1: Confirm quota ownership and high-priority exception governance.

Traceability: NCIE-002 NFRs; NCIE-004 engineering; NCIE-009 abuse controls.

Integration Security & Zero-Trust Enforcement

Bind NCIE-009 authentication, authorization, scoped machine identities, Tool/connector security and DLP requirements into every API/event surface.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-009 Ch.5-6, 16 and downstream handover.

27.0 Salient Things to Note

API ACCESS ≠ AUTHORIZATION. External/internal integration must be explicitly authenticated/authorized and constrained by purpose, classification and current policy — Table 34 binds every caller principal class in this document to a current-authorization check, with no principal class exempted by default.

27.1 Access Is Not Authorization

API ACCESS ≠ AUTHORIZATION, and AUTHENTICATION SUCCESS ≠ ACTION AUTHORITY. Every external and internal integration surface is explicitly authenticated and authorized, constrained by purpose, classification and current policy — no interface in this document is exempt from NCIE-009 enforcement merely because it is internal or because a caller previously authenticated successfully.

27.2 API Authorization Matrix (directive item 55)

The matrix below maps caller principal classes to target scope, authorization basis and audit requirement, so API access can never substitute for an authorization decision.

Table 34. API Authorization Matrix.

Caller Principal Class	Target Service/Object/Action	Purpose/Classification Basis	Current Authorization Decision/Enforcement	Audit Requirement
Human Principal (via ARGUS/UX)	Any interface in this document	NCIE-003 classification + purpose	NCIE-009 IAM current-authorization check at each call	NCIE-010 Ch.20 access audit
Agent Runtime identity	Agent Factory/Runtime, Model/Tool Gateway, Memory/Context APIs	Task Contract scope + effective principal	NCIE-009 Agent-identity check + NCIE-007 current-authorization propagation	NCIE-010 Ch.20 access audit
Service identity (internal)	Internal service APIs, Event bus	Service-to-service contract scope	NCIE-009 service-to-service authentication/authorization	NCIE-010 Ch.20 access audit
Partner machine identity	External partner APIs, webhook subscriptions	Partner contract scope, narrowly bound	NCIE-009 external-access authorization, scoped credential	NCIE-010 Ch.20/21 access + governance audit
Operator identity	Operator/regulatory interfaces	Governed submission/retrieval scope	NCIE-009 IAM + Master Register scope check	NCIE-010 Ch.20/21 access + governance audit
Validator identity	VPF service interfaces	Validation-request scope only	NCIE-009 validator-identity check	NCIE-010 Ch.20 access audit

27.R RACI / Control Ownership

Table 35. Chapter 27 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve high-risk API surfaces/additional security controls	Accountable Authority Class (to be assigned)	Integration Security Owner (to be assigned)	NCIE-009 Zero-Trust owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR11-27-1: Confirm high-risk API surfaces and additional security controls.

Traceability: NCIE-009 Ch.5-6, 16 and downstream handover.

Contract Observability, Audit & SLO Interface

Define observability requirements for APIs/events/connectors without conflating telemetry, audit and provenance.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-004 observability; NCIE-009 logging; NCIE-010 audit/provenance.

28.0 Salient Things to Note

Operational metrics do not prove business correctness; audit/provenance records remain distinct from traces and logs — this chapter never allows a clean dashboard to be cited as evidence a contract behaved correctly.

28.1 Metrics Do Not Prove Correctness

Operational metrics (latency, errors, saturation, retries, connector health, consumer lag, traces) do not prove business correctness; audit and provenance records (NCIE-010 Ch.19-21) remain distinct from traces and logs and are never inferred from them.

28.2 SLO Ownership

Every interface's observability surface links to its audit references. Interface SLO ownership and alert-escalation routing are Human Review decisions; no numeric SLO target is invented here.

28.R RACI / Control Ownership

Table 36. Chapter 28 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve interface	Accountable	Contract	NCIE-010	Assurance/Audit
SLO ownership/alert escalation	Authority Class (to be assigned)	Observability Owner (to be assigned)	audit/provenance owner	(NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Non-Blocking) HR11-28-1: Confirm interface SLO ownership and alert escalation; does not block the observability fields already defined in the interim.

Traceability: NCIE-004 observability; NCIE-009 logging; NCIE-010 audit/provenance.

Degraded Integration, Buffering & Reconciliation

Define behavior when event bus, partner systems, connectors or downstream services are unavailable.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.25.7; NCIE-004 resilience; NCIE-010 recovery.

29.0 Salient Things to Note

No event is silently dropped. Buffering/retry does not authorize infinite replay or side-effect duplication; reconciliation must surface gaps and uncertainty — this chapter's recovery-status output is required to name every unresolved gap explicitly, not merely report success.

29.1 No Silent Loss, No Blind Replay

No event is silently dropped during degradation — buffering, dead-letter/quarantine handling and eventual retry or replay preserve the record. Buffering/retry does not authorize infinite replay or side-effect duplication; reconciliation must surface gaps and uncertainty rather than assume completeness.

29.2 Recovery Thresholds

Gap detection, partial-availability handling and recovery status are explicit outputs of reconciliation (aligned with NCIE-010 Ch.26). Maximum buffering/replay tolerances and manual-intervention thresholds are Human Review decisions, not invented defaults.

29.R RACI / Control Ownership

Table 37. Chapter 29 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve buffering/replay tolerances and intervention thresholds	Accountable Authority Class (to be assigned)	Reconciliation Owner (to be assigned)	NCIE-010 recovery owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR11-29-1: Confirm maximum buffering/replay tolerances and manual intervention thresholds.

Traceability: NCIE-002 Ch.25.7; NCIE-004 resilience; NCIE-010 recovery.

NCIE-011 — Chapter 30 Expansion

API / Integration Testing & Contract Verification

Define contract, compatibility, security, idempotency, replay, failure, reconciliation and connector test classes without duplicating NCIE-016.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 acceptance; NCIE-009 security tests; NCIE-016 downstream.

30.0 Salient Things to Note

TEST SPECIFIED ≠ TEST EXECUTED. Passing schema validation alone is insufficient; semantic, security and failure-path behavior must also be verified — Table 38 specifies eight test classes and every status field is marked to prevent specification from being mistaken for execution.

30.1 Test Classes

Eight test classes are specified: contract tests, consumer-compatibility tests, authorization tests, event-replay tests, duplicate-suppression tests, retry-safety tests, connector fault-injection tests, and external sandbox tests.

Table 38. API/integration test-class specification.

Test ID	Test Class	Verifies	Status
INT-T1	Contract Test	Ch.3: request/response and event payloads conform to the published Service Contract schema	TEST SPECIFIED
INT-T2	Consumer Compatibility Test	Ch.20: a schema change does not break existing consumers under the additive-first compatibility rule	TEST SPECIFIED
INT-T3	Authorization Test	Ch.27: every caller principal class in Table 34 is denied access absent current authorization	TEST SPECIFIED
INT-T4	Event Replay Test	Ch.18, 22: replayed events do not reexecute side effects and consumers reconcile current state correctly	TEST SPECIFIED
INT-T5	Duplicate Suppression Test	Ch.22: accidental duplicates are suppressed while legitimate repeated actions are preserved	TEST SPECIFIED
INT-T6	Retry Safety Test	Ch.22: retries with an idempotency key do not duplicate consequential side effects	TEST SPECIFIED
INT-T7	Connector Fault-Injection Test	Ch.15-17, 29: connector failures are quarantined or buffered without silent loss	TEST SPECIFIED
INT-T8	External Sandbox Test	Ch.6: partner integrations behave correctly against a	TEST SPECIFIED

sandboxed partner
environment before
production onboarding

30.2 Test-Status Integrity

Specifying a test is not executing it (TEST SPECIFIED ≠ TEST EXECUTED). Passing schema validation alone is insufficient — semantic, security and failure-path behavior must also be verified. For every test above: IMPLEMENTATION EVIDENCE AVAILABLE = NO; TEST EXECUTED = NO; TEST PASSED = N/A; OPERATIONAL/PARTNER ACCEPTANCE = PENDING. Detailed test-case authoring and execution are downstream in NCIE-016.

30.R RACI / Control Ownership

Table 39. Chapter 30 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Integration)	Consulted	Informed
Approve test acceptance authority/partner certification requirements	Accountable Authority Class (to be assigned)	Integration Test Owner (to be assigned)	Assurance/Audit (NCIE-008, independent per Ch.6 SoD)	NCIE-016 downstream owner

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR11-30-1: Confirm test acceptance authority and partner certification requirements.

Traceability: NCIE-002 acceptance; NCIE-009 security tests; NCIE-016 downstream.

Human Review Register & Open Integration Decisions

Consolidate every unresolved API/integration decision exposed across Chapters 1-30 and Chapter 32 into one register: deduplicated, consistently classified, each stating precisely what it blocks. No item here manufactures closure of a decision the approved upstream chain, the approved NCIE-008/009 governance and security layers, or the approved NCIE-011 Skeletal v0.1 does not itself settle.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-008/009 Human Review registers, plus interface/integration-specific unresolved items from Chapters 1-30.

31.0 Salient Things to Note

Deduplicate identical NCIE-008/009 governance/security questions; create HR11 items only for distinct interface/integration decisions. Blocking scope must be precise.

31.1 Purpose

This register exists so that no unresolved integration decision is silently settled by expansion prose, and so that no chapter's own Human Review Focus item is left outside the register it is meant to feed — a defect found and corrected in the companion NCIE-008 v1.0 document's initial build. Every item from Chapters 1-30 below was extracted programmatically from that chapter's own Human Review Focus section (§31.4). Chapter 31's own item (HR11-31-1) and Chapter 32's own item (HR11-32-1) are each defined once as a fixed value and rendered identically both here and in their own chapter's Human Review Focus bullet, so neither can be omitted from the register or drift from the bullet that also states it.

31.2 Summary

Table 40. Human Review Register summary (mechanically computed).

Classification	Count
ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED	6
INSTITUTIONAL CONFIRMATION REQUIRED	19
LEGAL/POLICY CONFIRMATION REQUIRED	1
RESOLVED BY UPSTREAM BASELINE	1
SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED	5
Total	32
— Blocking	24
— Non-Blocking	8

31.3 Complete Register

Table 41. Complete Human Review Register.

Review ID	Source Chapter	Question	Classification	Blocking Scope	Upstream Cross-Reference	Status
HR11-1-1	Chapter 1	Confirm NCIE-011 acceptance/sign-off authority and any unresolved overlap with NCIE-013	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm NCIE-011 acceptance/sign-off authority and any unresolved overlap with NCIE-013	NCIE-008 HR8-10-1 (acceptance/after sign-off authority) — same	Open

		domain-module APIs.		domain-module APIs.	institutional question, instantiated for NCIE-011; resolving HR8-10-1 for the NCIE-series pattern resolves this item too.	
HR11-2-1	Chapter 2	Approve any institution-specific integration governance rules not fixed upstream; does not block the doctrine already stated above in the interim.	INSTITUTIONAL CONFIRMATION REQUIRED (Non-Blocking)	Non-blocking — does not gate chapter-level integration operation	N/A — distinct interface/integration decision	Open
HR11-3-1	Chapter 3	Confirm the authority for publishing/deprecating service contracts.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm the authority for publishing/deprecating service contracts.	N/A — distinct interface/integration decision	Open
HR11-4-1	Chapter 4	Confirm any permitted internal bypass patterns, if any, and the approval authority for them.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm any permitted internal bypass patterns, if any, and the approval authority for them.	N/A — distinct interface/integration decision	Open
HR11-5-1	Chapter 5	Confirm internal API classes requiring stricter assurance or isolation.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm internal API classes requiring stricter assurance or isolation.	N/A — distinct interface/integration decision	Open
HR11-6-1	Chapter 6	Confirm partner categories, onboarding authority and restricted data classes.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm partner categories, onboarding authority and restricted data classes.	N/A — distinct interface/integration decision	Open
HR11-7-1	Chapter 7	Confirm operator classes and authority boundaries by domain.	LEGAL/POLICY CONFIRMATION REQUIRED (Blocking)	Confirm operator classes and authority boundaries by domain.	N/A — distinct interface/integration decision	Open
HR11-8-1	Chapter 8	Confirm which ARGUS-facing operations may be synchronous versus asynchronous; does not block	INSTITUTIONAL CONFIRMATION REQUIRED (Non-Blocking)	Non-blocking — does not gate chapter-level integration operation	N/A — distinct interface/integration decision	Open

		the interface contracts already defined in the interim.				
HR11-9-1	Chapter 9	Confirm write authorities and cross-context exception handling.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm write authorities and cross-context exception handling.	N/A — distinct interface/integration decision	Open
HR11-10-1	Chapter 10	Confirm which Agent lifecycle operations require explicit Human/Governance approval.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm which Agent lifecycle operations require explicit Human/Governance approval.	N/A — distinct interface/integration decision	Open
HR11-11-1	Chapter 11	Confirm caller-visible versus internal routing metadata; does not block the provenance fields already required in the interim.	INSTITUTIONAL CONFIRMATION REQUIRED (Non-Blocking)	Non-blocking — does not gate chapter-level integration operation	N/A — distinct interface/integration decision	Open
HR11-12-1	Chapter 12	Confirm high-risk Tool classes and approval-reference requirements.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm high-risk Tool classes and approval-reference requirements.	N/A — distinct interface/integration decision	Open
HR11-13-1	Chapter 13	Confirm which VPF operations are callable by which technical principals.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm which VPF operations are callable by which technical principals.	N/A — distinct interface/integration decision	Open
HR11-14-1	Chapter 14	Confirm mutability/appended-only expectations and privileged audit access.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm mutability/appended-only expectations and privileged audit access.	N/A — distinct interface/integration decision	Open
HR11-15-1	Chapter 15	Confirm connector classes requiring manual attestation or source-specific approval.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm connector classes requiring manual attestation or source-specific approval.	N/A — distinct interface/integration decision	Open
HR11-16-1	Chapter 16	Confirm which domain connectors are standard versus bespoke; does not block the reusable pattern already defined in	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Non-Blocking)	Non-blocking — does not gate chapter-level integration operation	N/A — distinct interface/integration decision	Open

		the interim.				
HR11-17-1	Chapter 17	Confirm accepted transfer classes/formats and size/volume thresholds.	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking)	Confirm accepted transfer classes/formats and size/volume thresholds.	N/A — distinct interface/integration decision	Open
HR11-18-1	Chapter 18	Confirm event-retention/replay policy and category-specific delivery expectations.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm event-retention/replay policy and category-specific delivery expectations.	N/A — distinct interface/integration decision	Open
HR11-19-1	Chapter 19	Approve any additional event categories and ownership; does not block the eight-category baseline already defined in the interim.	INSTITUTIONAL CONFIRMATION REQUIRED (Non-Blocking)	Non-blocking — does not gate chapter-level integration operation	N/A — distinct interface/integration decision	Open
HR11-20-1	Chapter 20	Confirm deprecation windows and compatibility authority.	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking)	Confirm deprecation windows and compatibility authority.	N/A — distinct interface/integration decision	Open
HR11-21-1	Chapter 21	Confirm cross-boundary trace propagation limits for privacy/security.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm cross-boundary trace propagation limits for privacy/security.	N/A — distinct interface/integration decision	Open
HR11-22-1	Chapter 22	Confirm which operation classes require mandatory idempotency keys.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm which operation classes require mandatory idempotency keys.	N/A — distinct interface/integration decision	Open
HR11-23-1	Chapter 23	Confirm status vocabulary extensions and cancellation guarantees; does not block the seven-state baseline already defined in Table 28.	INSTITUTIONAL CONFIRMATION REQUIRED (Non-Blocking)	Non-blocking — does not gate chapter-level integration operation	N/A — distinct interface/integration decision	Open
HR11-24-1	Chapter 24	Confirm allowed webhook consumers and retry/deactivation policy.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm allowed webhook consumers and retry/deactivation policy.	N/A — distinct interface/integration decision	Open
HR11	Chapter	Confirm public	SECURITY/SOVEREIGNTY	Confirm public	N/A — distinct	Open

-25-1	er 25	versus internal error-detail exposure.	CONFIRMATION REQUIRED (Blocking)	versus internal error-detail exposure.	interface/integration decision	
HR11-26-1	Chapter 26	Confirm quota ownership and high-priority exception governance.	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking)	Confirm quota ownership and high-priority exception governance.	N/A — distinct interface/integration decision	Open
HR11-27-1	Chapter 27	Confirm high-risk API surfaces and additional security controls.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm high-risk API surfaces and additional security controls.	N/A — distinct interface/integration decision	Open
HR11-28-1	Chapter 28	Confirm interface SLO ownership and alert escalation; does not block the observability fields already defined in the interim.	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Non-Blocking)	Non-blocking — does not gate chapter-level integration operation	N/A — distinct interface/integration decision	Open
HR11-29-1	Chapter 29	Confirm maximum buffering/replay tolerances and manual intervention thresholds.	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking)	Confirm maximum buffering/replay tolerances and manual intervention thresholds.	N/A — distinct interface/integration decision	Open
HR11-30-1	Chapter 30	Confirm test acceptance authority and partner certification requirements.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm test acceptance authority and partner certification requirements.	N/A — distinct interface/integration decision	Open
HR11-31-1	Chapter 31	Review this consolidated register and decide, for each Blocking item, whether it must be resolved before NCIE-011 v1.0 may be approved or may instead remain open under an interim design default.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Review this consolidated register and decide, for each Blocking item, whether it must be resolved before NCIE-011 v1.0 may be approved or may instead remain open under an interim design default.	N/A — distinct interface/integration decision	Open
HR11-32-1	Chapter 32	Approve the NCIE-011 skeletal structure and authorize controlled full expansion —	RESOLVED BY UPSTREAM BASELINE (Non-Blocking)	Non-blocking — does not gate chapter-level integration operation	N/A — distinct interface/integration decision	Resolved by upstream baseline

already satisfied by the approved NCIE-011 Skeletal v0.1 and the production directive that authorized this v1.0 expansion; preserved here as historical record, not a live open item.	e
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31.4 Mechanical Verification

30 of these 32 items were recomputed programmatically from the Human Review Focus bullet text of Chapters 1-30 rather than assembled by hand. The remaining 2 items are HR11-31-1 (this chapter's own item) and HR11-32-1 (Chapter 32's own item), each defined once and rendered identically in both this table and their own chapter's Human Review Focus bullet. 0 duplicate Review IDs; 0 orphan Human Review items — every Human Review ID appearing anywhere in this specification is reconciled into this register.

31.5 Upstream Deduplication (Directive Item 61)

One item, HR11-1-1, restates an institutional question NCIE-008 has already opened as HR8-10-1 (acceptance authorities for each NCIE-series document and chapter-level sign-off); Table 41 cross-references it rather than treating it as an independent open question, per directive item 61. No other NCIE-011 item was found to duplicate an existing NCIE-008/009 governance or security question — each remaining item concerns an interface/integration-specific decision that NCIE-008/009 do not themselves resolve.

31.6 Blocking Scope Discipline

Each Blocking item states exactly what it blocks — a specific threshold, ownership assignment, security control or acceptance authority — rather than implying the whole of NCIE-011 or the whole approved upstream chain is blocked. Non-Blocking items are explicitly proposed defaults or confirmations that do not gate the corresponding chapter's integration operation in the interim.

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR11-31-1: Review this consolidated register and decide, for each Blocking item, whether it must be resolved before NCIE-011 v1.0 may be approved or may instead remain open under an interim design default.

Traceability: All chapters above (mechanically reconciled register); NCIE-008/009 Human Review registers.

Traceability, Acceptance & Downstream Handover

Close traceability to the approved NCIE-001-010 foundation and the approved NCIE-011 Skeletal v0.1; define acceptance criteria and the downstream handover to NCIE-012/013/014/015/016.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: Approved NCIE-001-010 foundation; NCIE-011 Skeletal v0.1.

32.0 Salient Things to Note

Skeleton approval authorizes expansion only. API architecture approval does not prove implementation, partner readiness, connector availability or test execution — §32.4's acceptance gate states this explicitly rather than leaving it implied.

32.1 Traceability Matrix

Every chapter of this document traces to an approved upstream authority or to the approved NCIE-011 Skeletal v0.1 itself. No chapter states an interface/integration requirement without an identifiable source.

Table 42. Full traceability matrix.

NCIE-011 Chapter	Primary Upstream Authority	Relationship
Chapter 1	Master Controlled Register v2.0; NCIE-002 Ch.25; NCIE-003/005/009/010	Mandate/integration boundary
Chapter 2	NCIE-002; NCIE-008; NCIE-009	API & integration doctrine
Chapter 3	NCIE-002 Ch.25.5; NCIE-003	Service Contract Model & interface registry
Chapter 4	NCIE-002 Ch.25; NCIE-009 Ch.5-6	API Gateway & entry-point architecture
Chapter 5	NCIE-002 Ch.25; NCIE-009	Internal service APIs
Chapter 6	NCIE-002 Ch.25.6; NCIE-009	External partner APIs
Chapter 7	Master Register; NCIE-001; NCIE-009	Operator/regulatory interface contracts
Chapter 8	NCIE-005; NCIE-006; NCIE-007	ARGUS API boundary
Chapter 9	NCIE-006 v1.2; NCIE-009	Memory & Context service interfaces
Chapter 10	NCIE-007 Ch.11-16; NCIE-003; NCIE-009	Agent Factory & Agent Runtime APIs
Chapter 11	NCIE-007 Model Gateway; NCIE-009	Model Gateway APIs
Chapter 12	NCIE-007 Tool Gateway; NCIE-009	Tool Gateway & connector execution APIs
Chapter 13	NCIE-003; NCIE-007 Ch.26; NCIE-008/009	VPF service interfaces
Chapter 14	NCIE-010 Ch.3, 7, 19	Evidence, custody & audit APIs
Chapter 15	NCIE-001; NCIE-002; NCIE-010	Source acquisition & ingestion connectors
Chapter 16	NCIE-013 (downstream); NCIE-003	Domain data connectors
Chapter 17	NCIE-001; NCIE-003; NCIE-009	Bulk file & batch integration

Chapter 18	NCIE-002 Ch.25.2-25.5	Event bus architecture & event contract model
Chapter 19	NCIE-002; NCIE-003/005/007/009/010	Event categories & domain event taxonomy
Chapter 20	NCIE-002 Ch.25.4; NCIE-003	Schema registry & contract versioning
Chapter 21	NCIE-002; NCIE-003; NCIE-010	Correlation, causation & trace propagation
Chapter 22	NCIE-002; NCIE-007	Idempotency, duplicate suppression & safe retry
Chapter 23	NCIE-002; NCIE-004/007	Asynchronous operations & long-running workflows
Chapter 24	NCIE-002; NCIE-009	Webhook & callback interfaces
Chapter 25	NCIE-002; NCIE-005; NCIE-007	Error semantics & failure taxonomy
Chapter 26	NCIE-002 NFRs; NCIE-004; NCIE-009	Rate limits, quotas & backpressure
Chapter 27	NCIE-009 Ch.5-6, 16	Integration security & Zero-Trust enforcement
Chapter 28	NCIE-004; NCIE-009; NCIE-010	Contract observability, audit & SLO interface
Chapter 29	NCIE-002 Ch.25.7; NCIE-004; NCIE-010	Degraded integration, buffering & reconciliation
Chapter 30	NCIE-002; NCIE-009; NCIE-016 (downstream)	API/integration testing & contract verification
Chapter 31	All chapters above; NCIE-008/009 Human Review registers	Consolidated Human Review Register
Chapter 32	Approved NCIE-001-010; NCIE-011 Skeletal v0.1	Traceability/acceptance/handover

32.2 Chapter-Count and Salient-Note Mechanical Verification

A programmatic scan of this document confirms exactly 32 sequential 'NCIE-011 — Chapter N Expansion' labels, N = 1..32, each present exactly once, in order, with no gap, duplicate or unintended renaming (directive item 74). A second scan confirms an explicit '<N>.0 Salient Things to Note' heading present for all 32 chapters (directive item 75), rerun independently against this final document rather than assumed from the build script.

32.3 Skeleton-to-Full Fidelity Verification

Every approved Chapter Purpose, Salient Things to Note, Principal Component, Upstream Dependency/Authority and Human Review Focus from the approved NCIE-011 Skeletal v0.1 is substantively represented in the corresponding full chapter above (directive item 73); none is omitted, and none is conceptually inverted.

32.4 Acceptance Criteria and the Unresolved-Item Gate

Full approval of NCIE-011 v1.0 requires: (a) 32/32 chapter integrity confirmed; (b) 32/32 explicit Salient Things to Note subsections confirmed; (c) skeleton-to-full fidelity confirmed; (d) every Blocking Human Review item in Chapter 31 either resolved by an institutional decision or explicitly and knowingly deferred by the approving authority; (e) the §30.1 test suite specified with no test falsely marked executed or passed; and (f) three-artifact consistency (this document, the Change Register and the Verification Report) confirmed. Approval of the v0.1 skeleton authorized this expansion only — it did not itself approve any threshold, product/protocol choice or ownership assignment proposed above. API architecture approval under this chapter does not prove implementation, partner readiness, connector availability or test execution.

32.5 Downstream Handover Register

NCIE-011 does not preempt detailed downstream specification. The table below states what is handed over and what remains downstream.

Table 43. Downstream handover register.

Downstream Document	What NCIE-011 Hands Over
NCIE-012 (UX)	ARGUS-facing and error/async-state contracts (Ch.8, 23, 25) for detailed conversational/visual design; no UX is specified here.
NCIE-013 (domain-module business logic)	Domain connector transport/schema-mapping patterns (Ch.16) for functional-module business rules; no domain interpretation is specified here.
NCIE-014 (physical storage)	Logical event/audit retention interface requirements (Ch.18, 28); no physical storage system or database is selected here.
NCIE-015 (operations)	Integration operational responsibilities (RACI/control-ownership tables throughout) for handoff into operational runbooks; no runbook is specified here.
NCIE-016 (detailed testing)	The 8 API/integration test classes specified in §30.1 (Table 38); detailed test cases, execution and partner certification remain downstream in NCIE-016 — this document specifies the tests, not their execution.

32.6 Readiness and Final Disposition

NCIE-011 API, Integration & Data Connector Specification v1.0 — Full Expansion of Approved Skeletal v0.1 — is READY FOR HUMAN REVIEW. It is NOT marked Approved. The approved v0.1 skeleton remains historically preserved as the expansion architecture; every proposed threshold, product/protocol choice and ownership assignment deferred in Chapters 1-30 remains a proposal pending the Chapter 31 Human Review Register; every §30.1 test remains SPECIFIED, not executed, pending NCIE-016.

Human Review Focus

- [RESOLVED BY UPSTREAM BASELINE] (Non-Blocking) HR11-32-1: Approve the NCIE-011 skeletal structure and authorize controlled full expansion — already satisfied by the approved NCIE-011 Skeletal v0.1 and the production directive that authorized this v1.0 expansion; preserved here as historical record, not a live open item.

Traceability: NCIE-001-010 approved foundation; approved NCIE-011 Skeletal v0.1.